

PROJECT MAINTENANCE HANDBOOK

QuietMel Project Maintenance and Technical Handover

Architecture, data contracts, metrics, operations, and troubleshooting

A code-backed operating baseline for future maintainers

Code baseline: main / 0ffe2b3

Baseline date: 10 August 2026 (Australia/Sydney)

Production project: Vercel / quietmel

Audience: developers, maintainers, model owners, and deployment owners

How to use this handbook

Executive summary

QuietMel is a map-first, low-stimulation travel application for Melbourne. Its differentiator is not simply finding the shortest route: it combines current City of Melbourne pedestrian readings, historical forecast profiles stored in Supabase, and Google walking alternatives to rank routes by practical quietness. The current product has no account system and does not persist user identity, location, journeys, or Google Place text.

This handbook treats the code on main at commit 0ffe2b3 as the source of truth. Where the README differs from the current implementation, the discrepancy is recorded under Known Gaps and Technical Debt.

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1. Product purpose and current scope

QuietMel helps people who can be affected by crowding, noise, and complex environments understand nearby conditions and choose lower-stimulation places and walking routes. The map remains the primary surface; controls appear only when needed.

1.1 Implemented capabilities

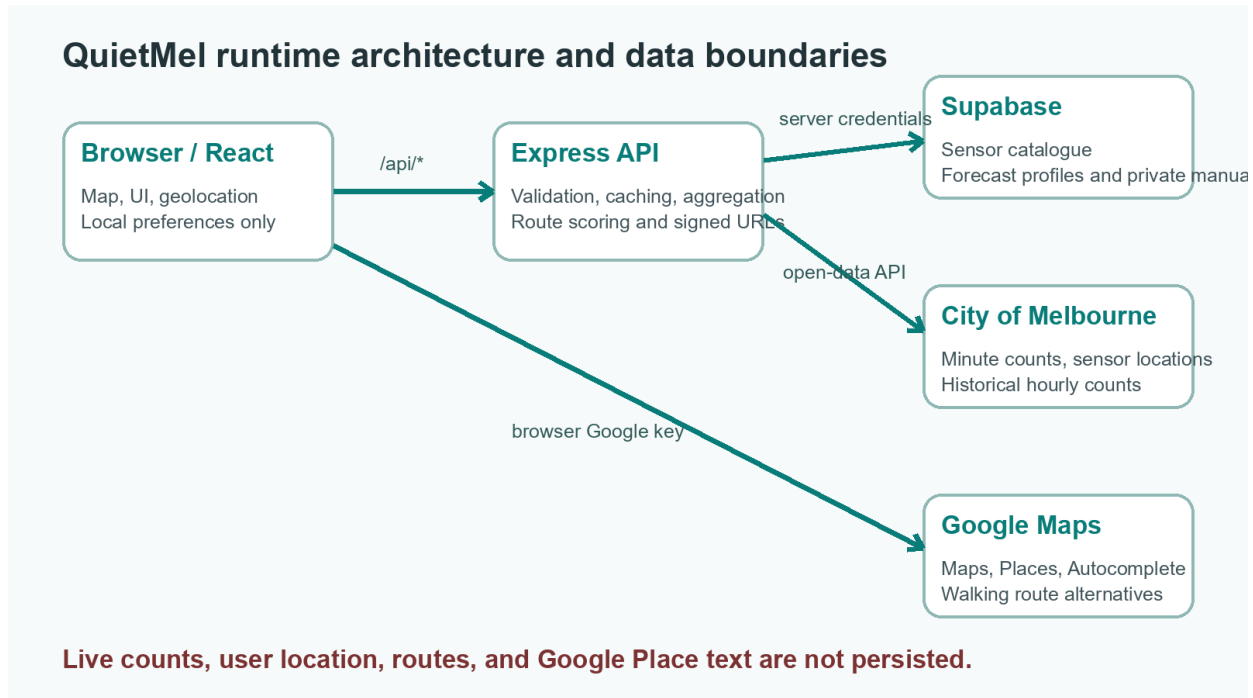
- Live pedestrian heatmap based on the latest available one-minute reading for each City of Melbourne sensor.

- Strict Heatmap, Sensors, and Combined map modes, including persistent sensor locations and live sensor detail cards.
- Automatic live refresh every 60 seconds and a manual no-cache refresh with visible loading feedback.
- A six-hour forecast rendered as 25 heatmap frames at 15-minute intervals, with play, pause, and timeline controls.
- Start and destination search using Google Autocomplete, with current-location or manually selected origins.
- Multiple Google walking alternatives ranked by a backend crowd-exposure score; selected routes are solid and alternatives are dashed.
- Nearby quiet-place discovery using semantic Google Place categories, distance ranking, and Google's fastest walking route.
- Eleven quick place categories for nearby discovery, with local browser preferences for visible categories.
- Location permission handling, current-position display, navigation-time position tracking, and blocking error feedback when permission is unavailable.
- Original and Dark themes, a language preference, a five-step first-run tour, and a backend-served PDF help centre.

1.2 Deliberate non-goals

- There is no current account, login, user profile, or server-side preference table. Earlier authentication and Turnstile code was removed.
- User location, searches, route paths, navigation progress, Google Place text, and minute-level live counts are not persisted.
- Navigation is not a full turn-by-turn engine: it displays the route and first instruction and updates the user marker, but has no off-route detection or voice guidance.
- Nearby quiet-place discovery uses place category and proximity as a proxy; it does not measure sound and currently does not filter candidate places by live crowd readings.

2. Runtime architecture and data boundaries



2.1 Request flow

- Browser to static app: React renders the map at / and HelpLibrary at /help.
- Browser to /api/*: Vercel rewrites the request to api/index.ts, which exports the shared Express app from server/app.ts.
- Express to City of Melbourne: the service requests live counts and active sensor locations with a 10-second source timeout.
- Express to Supabase: the server-only secret/service-role client reads and updates tables and creates private manual signed URLs.
- Browser to Google Maps Platform: Maps JavaScript, Places, Autocomplete, and Routes run directly in the browser with a referrer-restricted VITE key.

2.2 Persistence matrix

Data	Persisted?	Location / lifetime	Rule
Stable sensor locations	Yes	Supabase pedestrian_sensors	Used when the upstream location feed is unavailable
Minute live counts	No	City API plus 60-second Express memory cache	Avoids high-frequency database writes

Data	Persisted?	Location / lifetime	Rule
Historical forecast profiles	Yes	Supabase crowd_hourly_profiles	Only runtime forecast source
Google Place text	No	Current page session	Only optional google_place_id may be stored
Help PDF files	Yes	Private Supabase Storage	Opened with one-hour signed URLs
Theme/language/tour/categories	Local only	localStorage	No cross-device sync without accounts
User position and routes	No	React state and browser geolocation	Privacy-preserving, session-only

3. Technology stack and repository map

3.1 Technology stack

Layer	Technology	Version / purpose
Frontend	React and React DOM	19.2.8; UI and application state
Build	Vite	8.2.0; local proxy and production bundle
Language	TypeScript	6.0.3; strict frontend and backend checks
Maps	@vis.gl/react-google-maps	1.9.0; Google map, marker, and InfoWindow wrappers
Backend	Express	5.2.1; REST API and Vercel Function
Data platform	Supabase / PostgreSQL	Tables, RPC, RLS, and private Storage
Validation	Zod	4.4.3; quiet-route request contract
Icons	lucide-react	1.28.0; UI icon system
Model training	Python, pandas, numpy	Offline hourly-gradient-v1 profile generation
Hosting	Vercel	Vite frontend plus Express serverless entry

Dependency note

styled-components is installed in package.json, but the current interface is primarily styled with src/styles/*.css. Search for active imports before removing the package.

3.2 Repository responsibilities

Path	Responsibility	Maintenance note
src/App.tsx	Top-level state machine and mode transitions	Entry point for location, forecast, routes, settings, and tour
src/components/MapView.tsx	Google map and all overlays	Largest file; heatmap, places, routes, sensors, and location
src/components/RouteSearchPanel.tsx	Origin/destination input and autocomplete	Contains Safari blur/click workaround
src/components/RoutePlanner.tsx	Route alternatives and displayed metrics	Computes the displayed overall score
src/hooks	API request lifecycles	AbortController, duplicate suppression, and refresh timing
src/lib	Frontend contracts and API clients	crowd.ts and quietRoute.ts define key browser data shapes
server/routes	HTTP validation, headers, and error mapping	Keep business logic in services
server/services	Aggregation and domain logic	crowd, forecast, quietRoute, helpManuals
supabase/migrations	Database source of truth	Run in chronological order and version every schema change
model	Offline training and output artifacts	Excluded from Vercel and never read at runtime
scripts	Forecast import and manual upload	Loads server/.env, .env.local, then .env
api/index.ts	Vercel Express entry	Must remain consistent with vercel.json rewrites
docs	Team and maintenance documentation	Includes this handbook and history

4. Frontend application logic

4.1 Routing

src/main.tsx does not use React Router. It normalises window.location.pathname and renders HelpLibrary only for /help; every other path renders App. Vercel's SPA rewrite returns index.html for deep links. If the app gains more pages, adopt a proper router rather than adding many conditional branches.

4.2 Top-level application state

State	Default	Purpose and mode relationship
activeCategory	null	Non-null enters Google place discovery and hides crowd layers
crowdLayerMode	heatmap	heatmap / sensors / combined; mode changes clear sensor selection
forecasting	false	Exits place and route modes and forces heatmap
routePlanningActive	false	Displays scored route overlays and the route sheet
navigationActive	false	Shows only the selected route and starts watchPosition
userPosition	null	Memory-only browser coordinate
theme	original	Stored as quietmel:theme
language	en	Stored as quietmel:language
tourOpen	first visit	Controlled by quietmel:onboarding-complete

4.3 Layer separation and sensor details

- Heatmap mode renders only the canvas layer; no sensor marker exists to trigger a sensor InfoWindow.
- Sensors mode renders only AdvancedMarker elements and the selected InfoWindow. Stored locations remain visible without a current reading.
- Combined mode renders the heatmap first, then markers and InfoWindow. Route overlays use higher z-index values than the heat layer.
- The sensor card shows location name, pedestrians per minute, and reading time. Google detail text is cached per sensor for the current page session.
- If google_place_id exists, details are fetched by ID; otherwise a text query is biased to a 180-metre radius around the sensor.
- The card closes immediately from its button or 220 ms after pointer leave; pointer re-entry cancels the timer.

4.4 Place and route entry points

Entry	Search radius	Ranking / limit	Route behaviour
Quick place category	2 km from current location	DISTANCE / 20	Google's first walking route after place selection
Find quiet nearby	4 km from current location	DISTANCE / 20	First valid semantic place, then Google's fastest route

Entry	Search radius	Ranking / limit	Route behaviour
Origin/destination search	Australia with 50 km location bias	Autocomplete / 5 suggestions	Google alternatives, then Express crowd ranking

4.5 Location permission behaviour

- locateRequest starts at 1, so the map requests geolocation when Google Map becomes ready.
- Single-location requests use enableHighAccuracy=true and an 8-second timeout. Nearby quiet discovery also permits a 30-second cached position.
- Navigation uses watchPosition with maximumAge=5 seconds and timeout=10 seconds, and clears the watcher on cleanup.
- Quick place categories and nearby quiet discovery request permission if no current position exists and stop when permission is denied.
- The general route search intentionally still allows a manually selected origin when current location is unavailable.

4.6 localStorage contract

Key	Value	Effect when removed
quietmel:theme	original dark	Returns to Original theme
quietmel:language	en zh-CN	Returns to English
quietmel:onboarding-complete	true	Tour runs again on the next app load
quietmel.map-preferences	version 1 category preference object	Resets to parks, libraries, and cafes

5. Backend APIs and service behaviour

5.1 Endpoint catalogue

Method and path	Purpose	Caching / status	Primary failure
GET /api/health	Configuration check	200 or 503	Missing Supabase variables
GET /api/health/database	health_check RPC	No explicit cache	503 / PGRST202
GET /api/crowd/live	Current crowd snapshot	30s browser, 60s CDN, SWR 300s	502 source unavailable

Method and path	Purpose	Caching / status	Primary failure
GET /api/crowd/live?refresh=1	Forced live refresh	no-store	502 / 500
GET /api/crowd/sensors	Stable sensor catalogue	300s browser, 6h CDN, SWR 1d	502 / 500
GET /api/crowd/forecast	Six-hour forecast	no-store	503 model unavailable
POST /api/routes/quiet	Score route candidates	no-store	400 invalid, 503 forecast, 502 other
GET /api/help/manuals	Published manual catalogue	60s browser, 300s CDN	503 database
GET /api/help/manuals/:id/access	Private PDF signed URL	private 300s	404 / 503

5.2 Quiet-route request limits

- departureTime must be an ISO datetime.
- The request accepts one to four candidates. Candidate IDs are 1-40 characters.
- durationMinutes must be positive and at most 1,440; distanceMeters must be positive and at most 1,000,000.
- Each path contains 2-600 validated coordinates. The browser compacts route geometry to about 60 points before submission.
- Express limits JSON request bodies to 500 KB.

5.3 Response data contracts

Object	Important fields	Definition
LiveCrowdSnapshot	fetchAt, newestReadingAt, stale, pointCount, points, source	One latest validated point per sensor
LiveCrowdPoint	sensorId, name, lat/lng, pedestriansPerMinute, crowdLevel, intensity, measuredAt	Current map point
PedestrianSensor	sensorId, description, lat/lng, status, googlePlaceId	Stable location even when no live count exists
CrowdForecastSnapshot	startsAt, endsAt, horizonHours, intervalMinutes, sensors, frames	25 forecast frames for six hours
CrowdForecastValue	sensorId, PPM, level, intensity, qualityFlag, sampleCount	One sensor in one time frame
QuietRouteScore	average/maximum PPM, exposure, highCrowdPercent, coverage, combinedCost	Server-computed route metrics

5.4 CORS and generic errors

Production accepts origins listed in CLIENT_ORIGINS. Development also permits localhost:5173 and 127.0.0.1:5173. Requests without an Origin header are allowed. Unmatched routes return {error: 'not_found'} with 404; uncaught errors return {error: 'request_failed', message} with 500.

6. Data sources, database schema, and field definitions

6.1 External sources

Provider	Dataset / API	Use	Licence / storage
City of Melbourne	pedestrian-counting-system-past-hour-counts-per-minute	Current one-minute counts	CC BY 4.0; not stored
City of Melbourne	pedestrian-counting-system-sensor-locations	Name, coordinates, active status	CC BY 4.0; stable catalogue stored
City of Melbourne	pedestrian-counting-system-monthly-counts-per-hour	Two-year model training	CC BY 4.0; compact profiles stored
Google Maps Platform	Maps JavaScript, Places, Routes	Map, places, autocomplete, routes	Session-only Place text

6.2 pedestrian_sensors

Field	Type	Definition
location_id	bigint primary key	Stable City of Melbourne sensor identifier
sensor_name	nullable text	Upstream short name
sensor_description	text	Display name and descriptive fallback
latitude / longitude	double precision	Validated coordinates with a composite index
status	text	Defaults to A; current catalogue reads status A
google_place_id	nullable text	Optional persistable Google Place ID
source_name / source_url	text	Attribution and source endpoint
source_synced_at	timestampz	Latest upstream synchronisation
created_at / updated_at	timestampz	Record lifecycle timestamps

6.3 crowd_hourly_profiles

Field	Type / constraint	Business definition
location_id	bigint foreign key	Sensor identifier; profiles cascade on sensor deletion
day_of_week	0-6	0=Sunday through 6=Saturday
season	four values	Australian meteorological season
hour_of_day	0-23	Australia/Melbourne local hour
baseline_ppm	>=0	Median PPM for the group, or fallback statistic
trimmed_mean_ppm	>=0	Mean after removing the lowest and highest 10 percent
p25_ppm / p75_ppm	>=0	Linear-interpolated quartiles
gradient_ppm_per_hour	double	Next-hour baseline minus current baseline
sample_count	>=0	Direct group sample count, not replaced by fallback support
quality_flag	ok / low_sample / fallback	>=10 / 5-9 / less than 5 direct samples
source_start_date / end_date	date	Actual training window
model_version	text	Currently hourly-gradient-v1
imported_at	timestampz	Database import timestamp

Primary key: location_id + day_of_week + season + hour_of_day + model_version. Each sensor has 7 x 4 x 24 = 672 rows. The current 100-sensor model contains 67,200 rows.

6.4 help_manuals and private Storage

Field	Definition
id	Lowercase letters, numbers, and hyphens; stable API key
title_en / title_zh	Bilingual manual title
summary_en / summary_zh	Bilingual manual summary
storage_path	Unique object path inside the help-manuals bucket
page_count	Positive page count shown in the sidebar
sort_order	Catalogue order
is_published	Only true rows are exposed
updated_at	Metadata update timestamp

Row-level security

All three application tables enable RLS and revoke anon/authenticated access. Current reads and writes run through the server secret/service-role client. The help-manuals bucket is private, limited to 10 MB files and application/pdf MIME type.

7. Crowd, forecast, heatmap, and route metrics

7.1 Crowd level and intensity

```
crowdLevel(count):
  count < 50  => low
  count < 150 => medium
  otherwise   => high

intensity(count):
  count <= 50  => round(count / 50 * 35)
  count <= 150 => round(35 + (count - 50) / 100 * 40)
  count > 150  => round(75 + min((count - 150) / 150, 1) * 25)
```

Intensity is bounded from 0 to 100 and saturates at 300 pedestrians per minute. The high-crowd route threshold uses the same 150 PPM boundary as crowdLevel=high.

7.2 Heatmap rendering metrics

- Points with pedestriansPerMinute ≤ 0 or intensity ≤ 0 are not drawn. Zero explicitly represents no pedestrians.
- $\text{radius} = 36 + \text{intensity} * 0.3 + \text{pulse} * \text{pulseStrength} * 8$.
- $\text{pulseStrength} = \max(0, (\text{intensity} - 75) / 25)$, so pulsing begins only above intensity 75.
- $\text{centreAlpha} = 0.14 + \text{intensity}/100 * 0.13 + \text{pulse} * \text{pulseStrength} * 0.045$.
- Colour is linearly interpolated across five configured colour stops. Device pixel ratio is capped at 2.
- Continuous animation is throttled to at least 32 ms between paints and stops when reduced motion is requested or the document is hidden.

7.3 Forecast training and runtime formula

Item	Current baseline
Training window	7 Aug 2024 to 6 Aug 2026
Rows	1,614,110 input and clean rows; no duplicate groups
Sensors	103 source sensors; 28, 65, and 78 excluded; 100 trained

Item	Current baseline
Profiles	67,200 rows: 64,381 ok, 1,327 low_sample, 1,492 fallback
Validation	Latest 28-day temporal holdout; 99 sensors with test observations
Validation summary	Mean MAE 1.0402 PPM; median MAE 0.8285; median WAPE 17.0181%

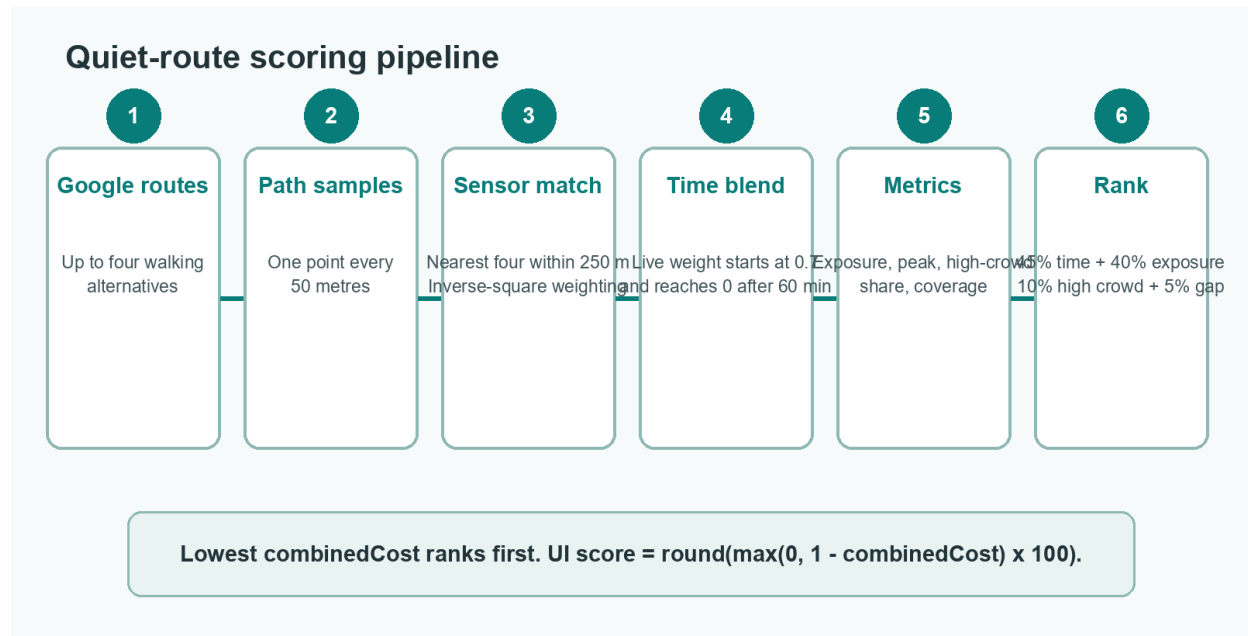
```
prediction = round_to_1_decimal(max(0, baseline_ppm + gradient_ppm_per_hour * minute / 60))
```

The service resolves weekday, Australian season, hour, and minute in Australia/Melbourne time. It creates 25 frames from now through +6 hours at 15-minute intervals.

7.4 Low-sample fallback order

1. Same sensor, weekday, and hour, ignoring season.
2. Same sensor, weekday/weekend grouping, and hour, ignoring exact weekday and season.
3. Same sensor and hour across all available dates.
4. If no samples exist, numeric fields stay zero, quality_flag=fallback, and gradient is zero.

7.5 Quiet-route scoring



```

liveWeight = max(0, 0.7 * (1 - arrivalMinutes / 60))
blended = live * liveWeight + forecast * (1 - liveWeight)
sensorWeight = 1 / max(40, distanceMetres)^2
routeCrowd = sum(blended * sensorWeight) / sum(sensorWeight)
exposure = sum(routeCrowd * minutesPerSample)
coverageConfidence = coveredSamples / totalSamples
  
```

A sample is covered when at least one forecast sensor is within 250 metres. Up to four nearest sensors are used. If none are nearby, crowd falls back to the frame-wide forecast average, but that sample does not increase coverageConfidence. The forecast frame is selected by rounding arrivalMinutes / 15 and clamping the result.

```
combinedCost =
  normalize(durationMinutes) * 0.45
+ normalize(crowdExposure) * 0.40
+ min(1, highCrowdPercent / 100) * 0.10
+ (1 - coverageConfidence) * 0.05

UI overall score = round(max(0, 1 - combinedCost) * 100)
```

Displayed metric	Definition
averageCrowdPpm	crowdExposure / durationMinutes
maximumCrowdPpm	Maximum estimated PPM across route samples
highCrowdPercent	Share of route minutes estimated at 150 PPM or above
extraMinutesComparedWithFastest	Candidate duration minus fastest candidate duration
crowdReductionPercent	Non-negative exposure reduction relative to the fastest candidate
crowdLevel	Average PPM below 50=low, below 150=medium, otherwise high
priority	Ascending combinedCost; priority 1 is recommended

8. Caching, concurrency, and fallbacks

Layer / resource	Lifetime	Concurrency control	Fallback
Browser live hook	60-second interval	Single requestRef; abort on cleanup	Retains last successful snapshot and exposes error
Express liveCache	60 seconds	Shared liveRequest Promise	Returns stale=true when an older cache exists
Express sensorCache	6 hours	Parallel pagination	Reads pedestrian_sensors when source fails
HTTP live	30s / 60s / SWR 300s	CDN	Manual refresh is no-store
HTTP sensors	300s / 6h / SWR 1 day	CDN	Stable Supabase catalogue
Forecast	no-store	Parallel queries for unique time	No local CSV fallback; missing data returns

Layer / resource	Lifetime	Concurrency control	Fallback
		keys	503
Google sensor details	Page session	Map<sensorId, Promise>	Failed requests are removed so they can retry
Manual signed URL	1 hour	Cached per manual ID	Refreshes when less than 60 seconds remain

Why long sessions do not grow without bound

The live hook permits one active request, Express stores only one current snapshot and one sensor map, the canvas clears before each paint, and forecast state is reset on exit. Refreshes do not append an unlimited browser history of snapshots.

9. Local development, migrations, imports, and deployment

9.1 Environment variables

Variable	Location	Required	Purpose
VITE_GOOGLE_MAPS_API_KEY	.env.local / Vercel	Yes	Browser-visible key; restrict by referrer and API
VITE_GOOGLE_MAPS_MAP_ID	.env.local / Vercel	Recommended	DEMO_MAP_ID locally; production Map ID recommended
SUPABASE_URL	server/.env / Vercel	Yes	Supabase project URL
SUPABASE_PUBLISHABLE_KEY	server/.env / Vercel	Required by health check	SUPABASE_ANON_KEY is accepted as legacy fallback
SUPABASE_SECRET_KEY	server/.env / Vercel	Yes	Server only; SERVICE_ROLE_KEY accepted as fallback
CLIENT_ORIGINS	server/.env / Vercel	Recommended in production	Comma-separated CORS allowlist
PORT	server/.env	No	Defaults to 3000 locally

9.2 Local commands

```
npm install
npm run dev      # Vite 5173 + Express 3000
npm run lint
npm run check:server
npm run build
npm run preview
```

9.3 Supabase migration order

5. 20260805000100_health_check.sql
6. 20260805000200_pedestrian_sensors.sql
7. 20260808000100_crowd_hourly_profiles.sql
8. 20260810000100_help_manuals.sql

Run the SQL files in order through Supabase SQL Editor or an already configured Supabase CLI workflow. Then verify GET /api/health/database.

9.4 Data import commands

```
npm run forecast:import -- --dry-run
npm run forecast:import
npm run help:upload -- --dry-run
npm run help:upload
```

The forecast import enforces exactly 100 sensors, 672 rows per sensor, 67,200 total rows, one model version, finite numeric values, and the absence of sensors 28, 65, and 78. It deletes the previous rows for that model version before inserting batches of 500.

9.5 Vercel deployment

- The repository is linked to the existing Vercel project quietmel. The production URL is <https://quietmel.vercel.app>.
- A push to GitHub main can trigger production deployment; a linked workspace may also use `vercel deploy . --prod`.
- `vercel.json` must match `/api/:path* -> /api` before the catch-all `/:path* -> /index.html` rewrite.
- `.vercelignore` excludes `model/` and `output/`. Runtime forecasts and manuals come from Supabase.
- After deployment, test `/api/health`, `/api/health/database`, `/api/crowd/live`, `/api/crowd/forecast`, and `/api/help/manuals`.

10. Security, privacy, and third-party configuration

- Never expose `SUPABASE_SECRET_KEY` or `SERVICE_ROLE_KEY` through a `VITE_` variable. They belong only in `server/.env` or Vercel server environment variables.
- Restrict the Google browser key to approved HTTP referrers and to Maps JavaScript API, Places API, Places API (New), and Routes API.
- Express disables x-powered-by, limits JSON to 500 KB, and validates route candidates with Zod.

- The product has no account or profile data. Location and routes remain in browser memory and are not sent to Supabase.
- Help files are private. The browser receives a short-lived signed URL, never a storage admin credential.
- Preserve City of Melbourne attribution and the CC BY 4.0 licence link in the interface and documentation.

11. Release testing and troubleshooting

11.1 Minimum release checklist

- npm run lint and npm run build succeed.
- Both allow and deny geolocation paths are tested; denied permission must stop location-dependent features.
- Heatmap, Sensors, and Combined modes are separated and the InfoWindow stays above the heat layer.
- Manual refresh sends refresh=1 and automatic refresh never creates parallel live requests.
- Forecast returns 25 frames, supports Now/+3hr/+6hr, and does not draw zero readings.
- Autocomplete selection works in Safari; route alternatives switch between solid and dashed styling.
- The help page calls /api/help endpoints and makes no /help/*.pdf static requests.
- Original/Dark theme, language selection, and the first-run tour are tested on desktop and mobile.

11.2 Troubleshooting matrix

Symptom	Likely cause	Action
/api/* returns 404	Vercel rewrite or api/index.ts	Check rewrite order and Function build logs
Database returns 503/PGRST202	Migration or schema cache	Run health_check migration and refresh schema
Live data returns 502	City API timeout or changed fields	Inspect crowd service error and stale-cache fallback
Forecast returns 503	Profiles absent, version mismatch, or no sensor join	Verify hourly-gradient-v1 rows and sensor IDs
Route scoring fails	Forecast unavailable or invalid request	Test forecast first, then interpret 400/503/502
Google map is blank	Key, billing, API, or referrer restriction	Verify all four APIs and production domain
Safari suggestions cannot be selected	Input blur removes list before click	Keep onPointerDown preventDefault on suggestion buttons

Symptom	Likely cause	Action
Manual PDF does not open	Signed URL, Storage MIME, or browser PDF support	Test access endpoint and a Range request returning 206
Denied location feature still continues	Error shown without an early return	Ensure permission failure stops subsequent search or routing

12. Known gaps and maintainer handover checklist

12.1 Code and documentation discrepancies

README route constraint mismatch

README says routes more than 25% or ten minutes slower than the fastest route are rejected. The current `server/services/quietRoute.ts` does not implement that filter; it scores every Google candidate returned. Add code and tests if the product still requires this constraint.

Nearby quiet-place naming mismatch

`findNearbyQuietPlace` selects the nearest semantically quiet category and Google's fastest route. It does not query sound levels, test current crowd readings, or select a low-activity sensor area. Some `RoutePlanner` wording is therefore stronger than the implemented logic.

- Chinese mode is not a full-site localisation. Settings, help, and some buttons are bilingual, but many map and route strings remain hard-coded in English.
- `MapView.tsx` and `App.tsx` are large and should be separated into heatmap, place, location, and route modules with focused tests.
- The repository has no automated test suite or CI workflow. Current gates are TypeScript, ESLint, and manual browser testing.
- Health configuration requires a publishable key even though core runtime services use the admin client; reassess whether this should remain mandatory.
- `NavigationDemo`, `demoRoutes`, and other prototype assets remain. Confirm imports before deleting them.
- `vite.config.ts` still mentions MapLibre in a comment, although the actual map engine is Google Maps.
- Keep all source files UTF-8 and verify the Chinese option in the browser after any text edit.

12.2 First-day handover checklist

9. Pull main, compare HEAD with the 0ffe2b3 baseline, and read all newer commits.
10. Configure personal Google and Supabase credentials without copying or committing production secrets.
11. Run lint/build, start both services, and test health, live, sensors, forecast, routes, and help APIs.
12. Confirm the three application tables, health_check RPC, private help-manuals bucket, and RLS rules in Supabase.
13. Confirm Vercel root, build, output, rewrite, and environment-variable settings.
14. Complete location, autocomplete, multi-route, and help-PDF flows in both Chromium and Safari.
15. Resolve the documented product mismatches before extending route or quiet-place logic.

Appendix A: source-file index

Concern	Primary files
Top-level modes	src/App.tsx
Map, heatmap, sensors, places	src/components/MapView.tsx
Live data	server/services/crowd.ts; src/hooks/useLiveCrowd.ts
Forecast	model/train_crowd_profiles.py; server/services/forecast.ts
Route scoring	server/services/quietRoute.ts; src/lib/quietRoute.ts
Search and autocomplete	src/components/RouteSearchPanel.tsx
Help centre	server/services/helpManuals.ts; src/components/HelpLibrary.tsx
Database	supabase/migrations/*.sql
Production routing	api/index.ts; vercel.json; .vercelignore

Update this handbook whenever a data contract, metric, scoring weight, migration, or production routing rule changes.